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Problem no.01:

Following data refers to the yields of a completely Random experiment with 4 treatments.

| A | B | C | D |
|---|---|---|---|
| 25 | 30 | 23 | 27 |
| 30 | 34 | 25 | 25 |
| 22 | 37 | 22 | 23 |
| 24 | 36 | 26 | 24 |
| | 39 | 23 | |
| | 27 | 25 | |

Analyze the data

Solution: The additive model for CRD with unequal observation is -

$$Y_{ij} = \mu + \epsilon_i + E_{ij} \qquad i = 1, 2, 3, 4$$

$$j = 1, 2, \ldots n$$

Where,

$Y_{ij}$ = the observation of the ith treatment in the jth replicate

$\mu$ = General mean

$\epsilon_i$ = Effect due to the ith treatment

$E_{ij}$ = Random error components which are assumed to be normally independently distributed with 0 mean and

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**Assumptions:**

(i) Observation and errors are independently distributed.

(ii) Observation and errors confirm to normal distribution with the equal variance

(iii) Treatments and environmental effects are additive.

Hypothesis:

$H_0$ : All the treatments effects are same

$H_1$ : All the treatments effects are not same

Treatment total: $T_A = 101$, $T_B = 203$, $T_C = 141$, $T_D = 103$

$$\text{Total SS} = \sum_i \sum_j y_{ij}^2 - \frac{G^2}{N}$$

$$= 15669 - \frac{551^2}{20}$$

$$= 518.95$$

Where,

$$G = \sum_{i=1}^{4} \sum_{j=1}^{n} y_{ij} = y$$

$$N = \sum_{i=1}^{4} n_i = 20$$

$$\text{Treatment SS} = \frac{\sum_{i=1}^{4} T_i^2}{n_i} - \frac{G}{N}$$

$$= \frac{101^2}{4} + \frac{203^2}{6} + \frac{141^2}{6} + \frac{103^2}{4} - 1518.05$$

$$= 386.887$$

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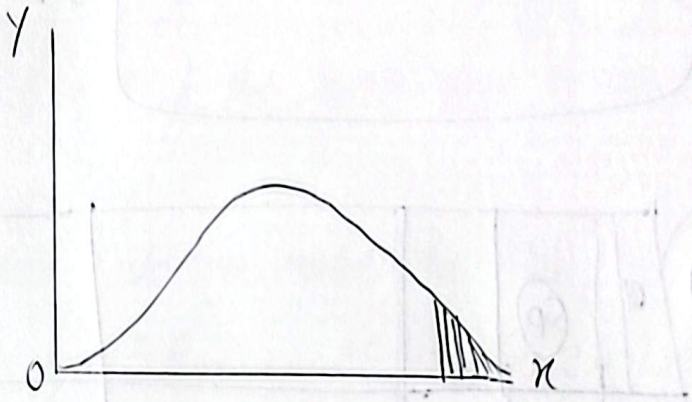
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$\therefore$ Error SS = Total SS - Treatment SS

= 518.95 - 386.887.

= 132.063

| Sources of variation | dif | SS | MS | $F_{cal}$ | $F_{tab}$ |
|---|---|---|---|---|---|
| Treatment | 3 | 386.887 | 128 % | $F_1 = 15.53$ | $F_{0.05,3,16} = 3.24$ |
| Error | 16 | 132.063 | 8.25 % | | |
| Total | 19 | | | | |




Comment: As the calculated value of F falls on the rejected region at 5% level of significance with 3 and 16 d.f.

So, the null hypothesis may be rejected. Hence we may conclude that all the treatments effect are not same.

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Problem-02:

The following data relate to a RBD with 4 treatment in 4 blocks.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | 18 | 20 | 20 | 21 |
| 2 | 17 | 19 | 19 | 20 |
| 3 | 16 | 18 | 17 | 20 |
| 4 | 15 | 17 | 16 | 18 |

Analyze the data and test the significance of treatments difference.

Solution: The linear additive model for RBD is

$$Y_{ij} = \mu + t_i + f_j + \epsilon_{ij}, \quad i = 1,2,3,4, \; j = 1,2,3,4$$

Where, $Y_{ij}$ = the treatment for ith treatment in jth block

$\mu$ = General mean

$t_i$ = Effect due to ith treatment

$t_j$ = Effect due to jth block

$\epsilon_{ij}$ = Random error components which are assumed to be normally distributed with 0 mean and variance $\sigma^2$

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**Assumption:**

(i) Observations and errors are independently distributed
(ii) Observations and errors are confirm to normal dist$^n$ with equal variance.
(iii) Treatment and environment effects are additive

**Hypothesis:**

① $H_{10}$: All the treatments effects are same.
$H_{11}$: All the treatments effects are not same

② $H_{20}$: All the blocks effect are same
$H_{21}$: All the blocks effect are not same.

Treatment total ($T_i$): $T_A = 66$, $T_B = 74$, $T_C = 72$, $T_D = 79$

Block total ($B_j$): $B_1 = 79$, $B_2 = 75$, $B_3 = 71$, $B_4 = 66$

$\therefore N = k \times n = 4 \times 4 = 16$

From calculation we get,
$\sum\sum y_{ij}^2 = 5339$
$\sum\sum y_{ij} = 291$

$$\text{Total SS} = \sum\sum y_{ij}^2 - \frac{G^2}{N}$$
$$= 5339 - \frac{291^2}{16}$$
$$= 46.44$$

$$\text{Treatment, SS} = \frac{T_i^2}{n} - \frac{G^2}{N}$$
$$= \left(\frac{66^2}{4} + \frac{74^2}{4} + \frac{72^2}{4} + \frac{79^2}{4}\right) - 5292.56$$
$$= 21.69$$

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$$\text{Block SS} = \frac{Bj^2}{K} - \frac{G^2}{N}$$

$$= \left(\frac{79^2}{4} + \frac{75^2}{4} + \frac{71^2}{4} + \frac{66^2}{4}\right) - 5292.56$$

$$= 23.19$$

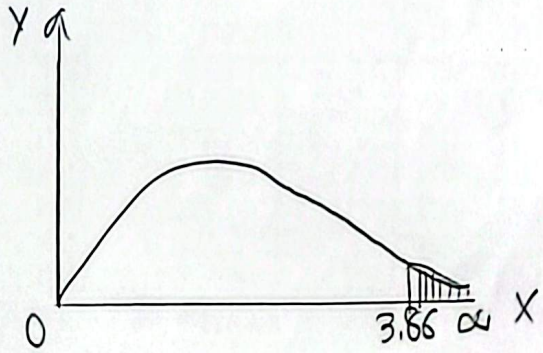
Error SS = Total SS - Treatment SS - Block SS

$$= 46.44 - 21.69 - 23.19$$

$$= 1.56$$

ANOVA TABLE

| Sources of variation | d.f. | SS | MS | $F_{cal}$ | $F_{tab}$ |
|---|---|---|---|---|---|
| Treatment | 3 | 21.69 | 7.23 | $F_A = 42.33$ | $F_{0.05,3,9} = 3.86$ |
| Block | 3 | 23.19 | 7.73 | $F_B = 45.47$ | $F_{0.05,3,9} = 3.86$ |
| Error | 9 | 1.56 | 0.173 | | |
| Total | 15 | | | | |

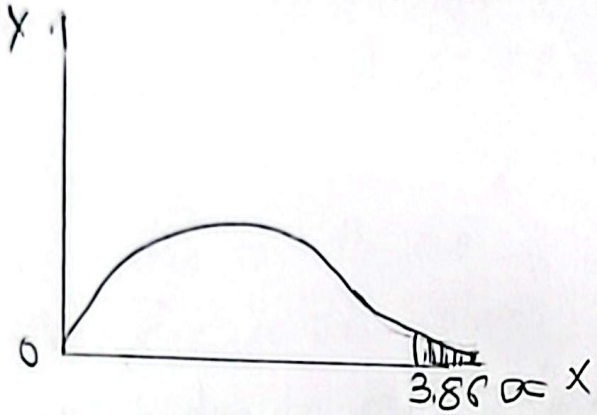



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Comment for $H_{10}$: As the calculated value of $F_A$ falls on the rejected region at 5% level of significance with 3, and 9 d.f so, the null hypothesis may be rejected. Hence we may conclude that all the treatments effects are not same.




Comment: As the calculated value of $F_B$ falls on the rejected region at 5% level of signifance with 3,9 d.f. So, we may reject the null hypothesis. Hence we may conclude that all the block effect are not same.

Problem no.03:

A 4×4 latin square was laid out to test the effects of various fertilizers on the yields of sugarcane in 4no acre plot. Given below is the field plan with plot yields.

| Row \ Column | | | | |
|---|---|---|---|---|
| | $A_{14}$ | $D_{22}$ | $B_{20}$ | $C_{18}$ |
| | $B_{19}$ | $C_{21}$ | $D_{16}$ | $A_{28}$ |
| | $C_{23}$ | $B_{13}$ | $A_{20}$ | $D_{18}$ |
| | $D_{21}$ | $A_{46}$ | $C_{24}$ | $B_{21}$ |

Perform analysis of variance and test the signifance of fertilizers effects.

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Solution:

The linear model of LSD is

$$Y_{ijs} = \mu + r_i + c_j + t_s + \epsilon_{ijs}$$

Where,

$Y_{ijs}$ = The observation of the sth treatment in the ith row and jth column ($i = 1,2,3,4$, $J = 1,2,3,4$, $S = 1,2,3..$)

$\mu$ = General mean

$r_i$ = The effect due to ith row

$c_j$ = The effect due to jth column

$t_s$ = The effect due to sth treatment

$\epsilon_{ijs}$ = The error components which are assumed to be independently and normally distributed with 0 mean and variance $\sigma^2$

Assumptions:

(i) Observations and errors are independently distributed

(ii) Observations and errors confirm to normal distribution with equal variance

(iii) Treatments and environment effects are additive

Hypothesis:

① $H_{10}$: All the rows effect are same

$H_{11}$: All the rows effect are not same.

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(ii) $H_{20}$: All the columns effect are same
$H_{21}$: All the columns effect are different

(iii) $H_{30}$: All the treatments effect are same.
$H_{31}$: All the treatments effect are not same

Here,
$N = V \times V = V^2 = 4^2 = 16$

$$\text{Total SS} = \sum_{i=1} \sum_{j=1} Y_{ijs}^2 - \frac{G_2^2}{N}$$
$$= 8043 - 7262.56$$
$$= 775.44$$

$T_A = 103$, $T_B = 75$, $T_C = 86$, $T_D = 77$

$$\text{Treatment SS} = \frac{T_s^2}{V} - \frac{G_2^2}{N}$$
$$= \left(\frac{103^2}{4} + \frac{75^2}{4} + \frac{86^2}{4} + \frac{77^2}{4}\right) - 7267.56$$
$$= 122.19$$

$R_1 = 74$ $R_2 = 79$, $R_3 = 76$, $R_4 = 112$

$$\text{Row SS} = \frac{R_1^2}{V} - \frac{G_2^2}{V}$$
$$= \left(\frac{74^2}{4} + \frac{79^2}{4} + \frac{76^2}{4} + \frac{112^2}{4}\right) - 7267.56$$
$$= 241.69$$

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$C_1 = 77$, $C_2 = 1041$, $C_3 = 80$, $C_4 = 80$

Column SS $= \frac{C_j^2}{r} - \frac{G^2}{N}$

$= \left(\frac{77^2}{4} + \frac{104^2}{4} + \frac{80^2}{4} + \frac{80^2}{4}\right) - 7267.56$

$= 118.69$

Error SS = Total SS − Treatment SS − Row SS − Column SS

$= 775.44 - 122.19 - 241.69 - 118.69$

$= 292.87$

ANOVA TABLE

| Sources of variation | d.f | SS | MS | Fcal | Ftab |
|---|---|---|---|---|---|
| Treatment | 3 | 122.19 | 40.73 | $F_1 = 0.82$ | $F_{0.05,3,6} = 4.76$ |
| Row | 3 | 241.69 | 80.56 | $F_2 = 1.65$ | $F_{0.05,3,6} = 4.76$ |
| Column | 3 | 118.69 | 39.56 | $F_3 = 0.81$ | $F_{0.05,3,6} = 4.76$ |
| Error | 6 | 292.87 | 48.81 | | |
| Total | 15 | | | | |

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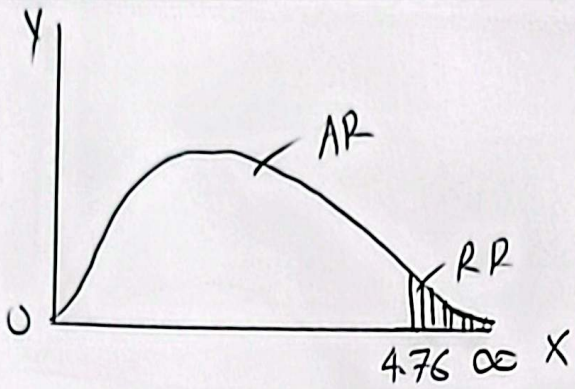
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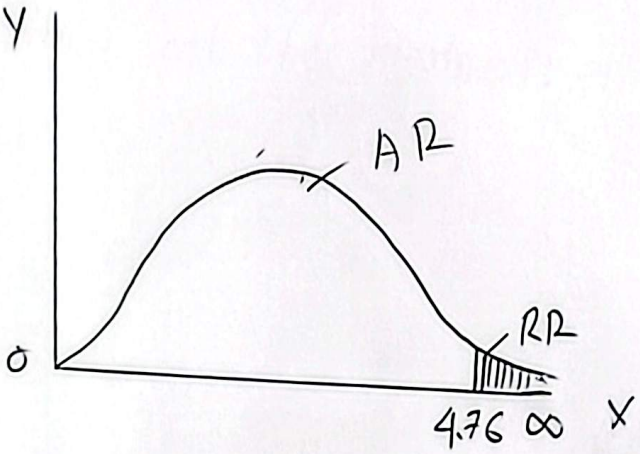
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Comment for $H_{10}$: As the calculated value of $F_1 = 0.82$ falls on the accepted region at 5% level of significance with 3 and 6 d.f. So, the null hypothesis may be accepted. Hence we may conclude that all the rows effect are same.




Comment for $H_{20}$: As the calculated value of $F_2 = 1.65$ falls on the accepted region at 5% level of significance with 3, 6 df. So, the null hypothesis may be accepted. Hence we may conclude that all the column effects are same.

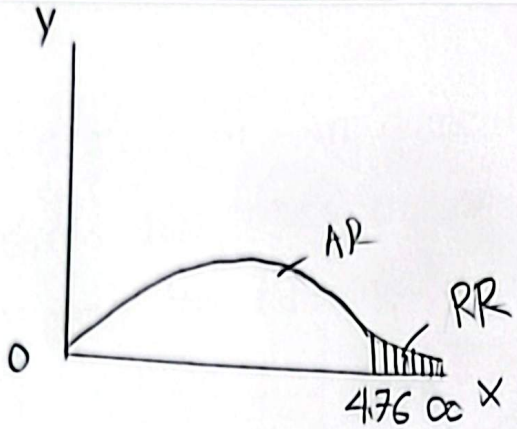
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Comment for $H_{30}$: As the calculated value of $F_3$ (0.81) falls on the accepted region at 5% level of significance with 3 and 6 d.f. so, the null hypothesis may be accepted. Hence, we may conclude that all the treatments effect are same

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